Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/20/2007
PI Name:	Chylack, Leo M.D.		
Project Title:	Precise Assessment of Prevalence and Progression of Lens Opacities in Astronauts as a Function of Radiation Exposure During Space Flight and Development of Improved Routine Clinical Assessment of Ocular Lens Status		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	leo@chylackinc.com	**Fax:**	FY 781-934-2747
PI Organization Type:	UNIVERSITY	**Phone:**	617-732-7355
Organization Name:	Harvard Medical School, Brigham & Women's Hospital		
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City:	Boston	**State:**	MA
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Comments:	If I am not available at 617-732-7355, my office number, please call 781-934-5052, my home number. [Ed. note: email changed to leo@chylackinc.com on 8/23/12]		
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	01/27/2003	**End Date:**	08/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 8/31/2008 per JSC (11/07) NOTE: End date changed back to 01/27/2008 per S. Krenek/JSC (8/07) End date changed to 3/31/2008 per JSC info update (10/06)		
Key Personnel Changes/Previous PI:	In July, 2006 after providing excellent optometric backup to the NASCA study the optometrists in Space Center Eye Associates left the project and were replaced by two new optometrists, Robert Gibson, O.D. and Jung Choi, O.D. The transition was a smooth one. Training in the standardized techniques of the routine eye examination for astronauts was provided, and training and certification in HIPAA and LOCS III classification were provided. The new optometrists have maintained the pace of the project. In the fall of 2006 for three months Cynthia Bell, M.S assisted Alan H. Feiveson, Ph.D. in the statistical analyses for the two NASCA manuscripts submitted to Archives of Ophthalmology in December, 2006. She then went onto a faculty position in academe. NASCA Administrative Assistant, Ms. Evelyn Hernandez left the BWH and the project in mid-2006 and was replaced by Ms. Nancy Leslie. This transition was also smooth and non-disruptive.		

COI Name (Institution):	Cucinotta, Francis (NASA Johnson Space Center) Feiveson, Al (NASA) Wear, Mary (Wyle Labs, NASA) Peterson, Leif (Resigned from Baylor College of Medicine in January 2007. Now at The Methodist Hospital in Houston, TX)
Grant/Contract No.:	NAG9-01491
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The NASA Study of Cataract in Astronauts (NASCA) is a cross-sectional and longitudinal five-year epidemiological study of the risk factors associated with cataract severity and progression in the whole population of American astronauts and two control populations – aircrew with military aviation experience and ground-based participants in the Longitudinal Study of Astronaut Health (LSAH). The study of risk factors focuses on the types and doses of radiation exposure in space flight, on measures of terrestrial solar UV radiation exposure, measures of nutrition, smoking, and general health. The assessment of cataract will use standardized digital images of the lens and validated measures of severity. The study will measure the severity and progression rates of cortical, nuclear, posterior subcapsular, and mixed cataracts, and it will relate these measures to measures of UV and ionizing radiation, nutrition, smoking, and general health. A secondary goal of this project will be to improve the routine annual clinical assessment of the ocular lens by including Nidek EAS 1000 digital imaging of the lens in the annual ocular examination protocol for astronauts. NASCA contains an initial cross-sectional assessment of the severity of the three types of lens opacification, and a follow-on assessment of progression rates of the three classes lens opacification in the populations of astronauts, military aircrew, and ground based comparison participants in the Longitudinal Study of Astronaut Health (LSAH). The main goal of the five-year longitudinal study is to determine the progression rates of the three main types of lens opacification in the complete sample of astronauts, the control populations of military aircrew and ground-based subjects in the LSAH, and then determine the risk factors associated with cataract progression with a specific focus on the components and doses of radiation exposure during space flight. Specifically, total radiation lens dose, space radiation lens dose, and individual contributions from space galactic cosmic ray and trapped proton lens dose will be assessed. NASA is also concerned about identifying and ultimately mitigating the risks to astronaut health of exposure to radiation in space. Several avenues of research now suggest that increased risk of lens opacification may be one of these adverse health effects. In order to address this issue, NASA has approved and funded this five-year, multi-centered research proposal entitled “The Precise Assessment of Prevalence and Progression of Lens Opacities in Astronauts as a Function of Radiation Exposure During Space Flight.” The Brigham and Women’s Hospital (BWH), the Johnson Space Center (JSC), the Departments of Medicine at Baylor College of Medicine (BCM) and The Methodist Hospital (TMH) both in Houston, Wyle Laboratories, and Space Center Eye Associates are the six centers cooperating in the execution of this study. The epidemiologic team at BCM and TMH will recruit astronauts and control subjects. Members of the Flight Medicine Clinic at the JSC and will perform ocular and general medical examinations. They will obtain specialized digital images of the crystalline ocular lens that will enable investigators at The Center for Ophthalmic Research (COR) at the BWH to derive measures of the severity for each class of cataract. Wyle Laboratories personnel at JSC will create and maintain the main data set of this project, and members of the Radiation Safety Office and the Statistical Branch of the JSC will work with Dr. Chylack, the PI of this project, and the other Co-Investigators to analyze of the data from this project and prepare regular reports and manuscripts. Leo T. Chylack, Jr., M.D. (BWH) originally submitted the proposal as a Supplemental Medical Objective (SMO). The goal of the proposal was to supplement the current annual ocular examination with new measurements that would allow an objective and more quantitative routine assessment of the status of the crystalline lens in astronauts. This supplementary methodology has enabled NASA to obtain objective assessments of the clarity of the lens, quantitative, continuous measures of the severity of lens opacification, and, over time, cataract type-specific progression rates. The SMO has undergone review by several intramural boards and a non-advocate peer review (NAR) panel. As a result of this review, the NAR recommended 1) a major expansion of the project to enable measures of severity and progression rates of various forms of lens opacification, and 2) modifications of the number and composition of the control groups to provide comparisons of severity and progressions rates of lens opacification in astronauts to those in military exposed to the cataractogenic risks of high altitude (but not space) flight and to those in LSAH subjects, who presumably are not exposed to high altitude or space radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Expanding our understanding of the mechanisms of space-radiation-induced cataract may suggest means of reducing the risk of radiation-induced cataract on earth among individuals employed in jobs in which radiation constitutes an occupational hazard. Also, it may suggest improved means of shielding the eyes of patients undergoing radiation therapy. Information about the mechanisms of cataracts in astronauts may suggest additional research into the causes of age-related cataract, the world’s leading cause of blindness. Lastly, the longitudinal phase of the study which links nutritional data to the risk of cataract progression may suggest nutritional means of ameliorating the risk of cataract.
	This Progress Report covers the period February 1, 2006 through January 31, 2007. Recruitment for NASCA began in July 2004. Our goal was to recruit all of the 285 US astronauts, 100 military aircrew members, and 100 ground-based controls. Most of our subjects were enrolled wihtin the first 18 months of active recruiting, a few were added in the subsequent six months. As of June 2006, we had enrolled 224 astronauts (78.6% of goal), 95 military aircrew (95% of goal), and 99 ground-based controls (99% of goal). The shortfall in enrolled astronauts was due largely to death, intervening cataract surgery (an exclusion criterion), and disinclincation to participate among older astronauts. The NASCA project is proceeding with these three cohorts. Over the summer of 2006 we completed the statistical analysis of the baseline data - the first year's data - and prepared two manuscripts: 1) describing the recruiting procedures and results, the methodologies used, the propensity scores for

Task Progress:

each cohort, and the variables likely to confound the statistical analysis, and 2) describing the association between space radiation exposure,solar UV radiation exposure, nutrition, smoking, general health factors, and medications used to the severity of cortical (C), nuclear (N), and posterior subcapsular (P) lens opacification.

In order to facilitate the abbreviation of these reports, we set up a second NASCA web site using a server that would hold all of the detailed materials about techniques, methods, policies, etc. and be accessible to the public. By removing all of the fine methodological detail from the two manuscripts and locating them on the publically accessible web site, we were able to meet the journal's requirements about manuscript length without limiting disclosure of the study's methods. The web site's URL is (http://nasca-study.bwh.harvard.edu).

A summary of the findings of the two manuscripts is provided below:

MANUSCRIPT 1:

TITLE: NASCA Report 1: Study of Relationship of Exposure to Space Radiation and Risk of Lens Opacity

AUTHORS: Leo T. Chylack, Jr.(a), M.D., Leif E. Peterson, Ph.D.(b), Alan H. Feiveson, Ph.D.(c), Mary Wear, Ph.D.(e), F. Keith Manuel, O.D.(e), William H. Tung (a), Dale Hardy, M.S.(b), Lisa Marak, R.N.(d), and Francis A. Cucinotta, Ph.D.(c)

AFFILIATIONS: a) Center for Ophthalmic Research, Brigham and Women's Hospital, Boston, MA; b) Department of Medicine , Baylor College of Medicine, Houston, TX; c) Lyndon B. Johnson Space Center, NASA, Houston, TX; d) Wyle Laboratories, Houston, TX; e) Space Center Eye Associates, Houston, TX.

ABSTRACT:

Objectives: The NASA Study of Cataract in Astronauts (NASCA) is a five-year investigation of lens opacification in the U.S. astronauts and two exposure control groups with and without a history of military aviation, respectively. The goals are to determine the effect of space radiation exposure on the severity and progression of age-related cortical (C), nuclear (N), and posterior subcapsular (PSC) lens opacities.

Methods: Astronauts and exposure controls were recruited from the local occupational aerospace community. Severity of C, N, PSC lens opacification, age, demographics and general health, nutritional intake, solar ocular exposure, and other confounding variables were measured at baseline. Results: A total of 419 subjects were enrolled (223 astronauts, 96 military aviator controls, and 100 ground controls) who met inclusion criteria. Significant confounder variables were a history of asthma, history of hypertension, history of obesity, number of drug reported taking, and maximum high contrast LogMAR acuity. Propensity scores based on confounders suggest that military aviator controls were more similar to astronauts when compared with ground controls.

Conclusions: Confounder patterns based on military aviator controls were more similar to astronauts when compared with ground controls. This will be important for dose-response models employed during the cross-sectional and longitudinal analyses.

MANUSCRIPT 2:

TITLE: NASCA Report 2 – Cross-sectional Analyses of Space Radiation Exposure and Risk of Lens Opacity

AUTHORS: Leo T. Chylack, Jr., M.D.(a), Alan H. Feiveson, Ph.D.(b), Leif Peterson, Ph.D.(c), F. Keith Manuel, O.D.(e), Mary L. Wear, Ph.D.(d), William H. Tung, B.S.(a), Dale Hardy, M.S.(c), Lisa Marak, R.N.(d), Cynthia Bell, M.S.(f), Francis Cucinotta, Ph.D.(b)

AFFILIATIONS: a. Center for Ophthalmic Research, Brigham and Women's Hospital, Boston, MA; b) NASA, Johnson Space Center; c) Baylor College of Medicine, Department of Medicine; d) Wyle Laboratories; e) Space Center Eye Associates; f) USRA Division of Space Life Sciences

ABSTRACT: Objectives: NASCA is a five-year investigation in U.S. astronauts to determine the effect of space radiation exposure on severity and progression of cortical (C), nuclear (N), and posterior subcapsular (PSC) lens opacities.

Methods: Objective measures of C, P, and N opacification were obtained from Nidek EAS 1000 digital lens images. Space radiation exposure was expressed as total lens dose/subject aggregated over missions. Solar ocular exposure and nutritional status were characterized. Statistical analyses consisted of fitting customized non-normal regression models to measures of opacity and testing effects of subject group, radiation exposure and other variables on degree of opacity.

Results: Age is the most important predictor of C. Older subjects had higher mean and more varied C. Astronauts who have not yet flown and aircrew controls had less C than controls, while astronauts who flew at least one mission in space had significantly greater C. Increased solar UV was associated with higher, and larger intake of A-vitamins with lower, C. Astronauts had less N, however neither space nor solar radiation exposure was associated with more N. No nutritional factors were associated N. Age was weakly associated with increased P.

Conclusions: Exposure to space radiation increases risk of C opacification.

As of February, 2007 the NASCA project is well into its third year, and when the last of the three-yearm evaluations is completed, we will be able to begin the first set of analyses of the longitudinal phase of the project in which we examine the relationship between the factors of interest to the rate of progression of lens opacification. This will be an important milestone in this project, since these measures are likely to be of greatest interest to NASA and the greater eye and radiation research communities.

Bibliography Type: Description: (Last Updated: 08/21/2012)

Abstracts for Journals and Proceedings	Chylack LT Jr, Feiveson AH, Peterson LE, Manuel FK, Wear ML, Hardy D, Marak L, Tung WH, Hernandez E, Cucinotta FA. "The NASA Study of Cataract in Astronauts (NASCA): Year-1 Data. " Annual Meeting of the Association for Research in Vision and Ophthalmology (ARVO), Ft. Lauderdale, FL, May 3, 2006. Invest Ophthalmol Vis Sci 2006 Apr;47: E-Abstract 4140. , Apr-2006
Abstracts for Journals and Proceedings	Chylack LT Jr, Feiveson AH, Peterson LE, Manuel FK, Wear ML, Tung WH, Hardy D, Marak L, Bell C, Cucinotta FA. "The NASCA study: cross-sectional analyses of exposure to radiation in space and risk of lens opacification." NASA Human Research Investigators' Workshop, League City, TX, February 12-13, 2007. NASA Human Research Investigators' Workshop, Feb 2007. , Feb-2007
Articles in Peer-reviewed Journals	Chylack LT Jr, Peterson LE, Feiveson AH, Wear ML, Manuel FK, Tung WH, Hardy D, Marak L, Cucinotta FA. "NASCA Report 1: Study of Relationship of Exposure to Space Radiation and Risk of Lens Opacity. " Submitted to Archives of Ophthalmology, December, 2006 (Now under revision for resubmission). , Feb-2007
Articles in Peer-reviewed Journals	Chylack LT Jr, Feiveson AH, Peterson LE, Manuel FK, Wear ML, Tung WH, Hardy D, Marak L, Bell C, Cucinotta FA. "NASCA Report 2 – Cross-sectional Analyses of Space Radiation Exposure and Risk of Lens Opacity. " Submitted to Archives of Ophthalmology, December 2006. (Now under revision for resubmission) , Feb-2007
Awards	Chylack LT Jr. "Invited to present an overview of the NASCA project at a Clinical Conference at the National Eye Institute at the National Institutes of Health, September 13, 2006." Sep-2006